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Dorothea Hofer, Dr. rer. nat., Dipl.-Phys.[1,2,3,5]
Jürgen Feldmeier, LL.M., Dipl.-Ing.[1,2,3,5]
Andreas Oser, LL.M., Dr. rer. nat., Dipl.-Chem.[1,2,3,5]
Susanne Sonnenhauser, Dr. rer. nat., Dipl.-Biol.[1,2,3,5]
Christian Gärtner, Dr.-Ing., Dipl.-Ing.[1,2,3,5]
Peter Klein, Dr. rer. nat., Dipl.-Phys.[1,2,3,5,7]
Bernhard Scheuvens, Dipl.-Ing.[2,7]
Christian Einsel, Dr. rer. nat., Dipl.-Phys.[1,2,3,5,7]
Alexander Lehmann, LL.M., Dipl.-Phys.[1,2,3,5,7]
Markus Adamczyk, Dipl.-Inf.[1,2,3,5,6,7]
Ulrike Henneböhle, Dr.-Ing., Dipl.-Ing.[2,7]
Sophia Steinmüller, B.Sc. Physics, M.Sc. Geophys.[2,7]
Ok Joo Lee, Dr. rer. nat., B.S., M.S.[4,7]

Rechtsanwälte / Attorneys-at-law [6]
Markus Adamczyk, Dipl.-Inf.[1,2,3,5,6,7]
Benjamin Ellrodt[6,7]
Marion Ziliox[6,7]

German Patent Attorney[1]
European Patent Attorney[2]
European Trade Mark and Design Attorney[3]
Korean Patent Attorney[4]
Registered Representative before the UPC[5]

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I.

The Applicant's desire to proceed further with the present application is herewith stated.

II.

On the Communication dated August 16, 2024, enclosing the Extended European Search Report and the Search Opinion (in the following shortly: EESR)

1. New documents

As per enclosure, a set of new claims 1 to 13 is filed – one fair copy and a marked-up working version to show the amendments.

It is respectfully requested to continue examination based on the amended set of claims.

2. Amendments and their admissibility (Art. 123(2) EPC)

New claim 1 is based on original claim 1, wherein however the term "or/and" has been replaced by "or". Accordingly, the option "and" has been deleted, meaning that the combination of measuring the amount of APOA2-AT protein and the amount of APOA2-OATQ protein has been deleted and thus is removed from the claimed subject-matter. Only the alternatives are now covered, i.e. either measuring the amount of APOA2-AT protein *or* measuring the amount of APOA2-ATQ protein. Such deletion, or removal of an embodiment (the combination) is allowable under Art. 123(2) EPC.

Reference is also made to the measurements and the results depicted from Figs. 4 and 5 of the original application: in the respective cases, the measurements were carried out with respect to APOA2-AT protein (cf. Fig. 4) or APOA2-ATQ protein (cf. Fig. 5) distinctively and separately, and consequently showing different distinct results on differentiation between benign and malign pancreatic cystic tumors, respectively.

Therefore, the provision of Art. 123(2) EPC is met.

In claim 9, the purpose for the determination is specified more clearly, in full consistency with the indication of claim 1, namely that the kit is for assisting in the determination of "*pancreatic cystic tumor of a test subject to be benign or malignant*". Further in line with the amendment in claim 1, the term "or/and" has been replaced by "or"; in addition the feature derived from original claim 10 has been incorporated into claim 9 (while at the same adapting the wording by replacing "or/and" by "or").

Then, original claims 11 to 14 follow as newly renumbered claims 10 to 13, respectively (considering proper new back references). Further, in new claim 12 as amended (derived from original claim 13), the wording "*specifically binds to an amino acid sequence other than the carboxyl-terminal region of*" is placed between the wordings "SEQ ID NO: 1 or" and "the APOA2-ATQ protein". This amendment is directly derivable from paragraphs [0027], [0028], [0057], [0101] and Fig. 5 of the original application.

Hence, the amendments comply with the requirements of Art. 123(2) EPC.

3. Clarity (Art. 84 EPC)

3.1 Independent claims 1 and 9

In **Item 5.1** and **5.2** of the EESR, it is questioned whether it is reliable and proper when using only the C-terminal region as the binding site for detecting ApoA2-ATQ and ApoA2-AT proteins. It is asserted that that the ApoA2-ATQ and ApoA2-AT proteins differ by only 1 amino acid and thus share many overlapping epitopes. Therefore, cross binding may happen and affect the measurement results.

However, particularly at p. 17, para.1, it is clearly stated that "*the anti-APOA2-AT terminus monoclonal antibody of the present invention specifically binds to the C-terminal region of the APOA2-AT protein shown in SEQ ID NO: 1, but exhibits no binding activity against the APOA2-ATQ protein shown in SEQ ID NO: 2 and the APOA2-A protein shown in SEQ ID NO: 3.*" The statement is further verified by Example 1 and Fig. 1 of present application which shows a specific binding of APOA2-AT protein without cross reactivity with ApoA2-ATQ protein. Figs. 4 and 5 then show distinctly discriminating results on the tumor being either benign or malignant, depending on whether the APOA2-AT-specific or the ApoA2-ATQ- specific sequences are used.

Therefore, it has been made credible and demonstrated that the C terminal region can serve as a binding site providing reliable measurement of the ApoA2-ATQ protein or the ApoA2-AT protein, even though the ApoA2-ATQ and ApoA2-AT proteins differ by only 1 amino acid.

3.2 Claim 8

In **Item 5.3** of the EESR, it is asserted that claim 8 comprises features that are contradictory with the subject matter of present claim 1.

However, this is respectfully disagreed to. The method defined in present claim 1 is to assist the determination of malignant pancreatic cystic tumor in a test subject having pancreatic cystic tumor. This underlined expression indicates that the test subject defined in present claim 1 is a subject that has been detected to have pancreatic cystic tumor.
Beyond this, claim 1 just further contributes by the determination of the pancreatic cystic tumor of the test subject to be benign or malignant. This is however by no means contradictory to claim 8.

Therefore, present claim 8 does not exist any features contradictory to present claim 1.

3.3 New claim 12 (original claim 13)

In **Item 5.3** of the EESR, it is asserted that it was not clear from original claim 13 whether anti-ApoA2 non-terminus monoclonal antibody does or does not bind to ApoA2-ATQ protein.

In order to address this clarity issue, the same wording of "specifically binds to an amino acid sequence other than the carboxyl-terminal region of" between the wordings "SEQ ID NO: 1 or" and "the APOA2-ATQ protein" has been incorporated into claim 13 (now claim 12).

Therefore, claim 12 as amended is clear in the sense of Art. 84 EPC.

4. Novelty

4.1 Before turning to a further discussion of the cited documents, it is indicated that document D1 (K. Honda et al., 2015) and D4 (WO 2015/050107 A1) have already been cited and discussed in the original application, notably as Non-Patent Literature 4 and as Patent Literature 2, respectively (see [0006] of the application as filed).

4.2 Indeed, it was known from the cited prior art that various pancreatic tumors including pancreatic cancer and pancreatic cystic tumor can be detected by measuring the amount of two types of variants which can form APOA2 protein dimers. The measurement includes measurement measuring *in vitro* the amount APOA2-AT protein **and** the amount of APOA2-ATQ protein. Accordingly, the methods described in D1 to D4 were merely directed to generally determine whether a patient has malignant pancreatic cystic tumor, as opposed to a healthy subject.

That is, D1 to D4 discloses combining the measurement of APOA2-AT and APOA2-ATQ, in order to determine whether a patient (compared to a healthy subject) has malignant pancreatic cystic tumor. However, none of D1 to D4 discloses, and therefore the subject-matter of claim 1 as amended differs from D1 to D4 in that

(1) The determination is directed to measuring the amount of APOA2-AT protein which is a polypeptide comprising the amino acid sequence represented by SEQ ID NO: 30 at the carboxyl terminus, **or** to measuring the amount of APOA2-ATQ protein which is a polypeptide comprising the amino acid sequence represented by SEQ ID NO: 31 at the carboxyl terminus, nor

(2) to make this measurement to specifically distinguish between the test subject **to be benign or malignant on the basis of the amount**.

Therefore, the subject-matter of present claim 1 as amended is novel over each of D1 to D4.

Due to the same or equivalent distinguishing features, the subject-matter of claim 9 is novel over each of D1 to D4 as well for the same reasons.

The subject-matter of original claim 14 (new claim 13) was not objected under novelty.

5. Inventive step (Art. 56 EPC)

5.1 Document D1 can be regarded as representing the closest prior art, see also "Non-Patent Literature 4" discussed in [0006] of the application as filed. It hence merely discloses that APOA2 protein variants may be useful for determining whether a patient has pancreatic cancer in comparison with healthy volunteers.

However, as described and explained in further detail in the background art (cf. in particular paragraphs [0002] and [0003] of the application as filed), it is difficult to determine whether a pancreatic cystic tumor is benign or malignant. Prior to the present invention, this differentiation had involved the performance of invasive methods. Therefore, it was a desire to provide non-invasive methods to determine whether a pancreatic cystic tumor (such as that of D1, or that described in D2 to D4) is benign or malignant.

5.2 As discussed in further detail in section 4 above, the subject-matter of present claims 1 and 9 differ from D1 (or D2 to D4) by distinguishing features (1) and (2).

Indeed, it has been surprisingly found that the concentration of the APOA2-**AT** protein in blood is lower in malignant pancreatic cystic tumor compared with benign pancreatic cystic tumor; and that the concentration of the APOA2-**ATQ** protein in blood is higher. These distinct different measurement results had been found and are demonstrated from the experimental part and specifically from Figs. 4 and 5, respectively. Reference is made also to the description and explanation in paragraph [0011] of the application as filed.

5.3 The claimed solution is not rendered obvious by any one of D1 to D4. That is, none of D1 to D4 disclose or suggest the surprising finding that determining and measuring *in vitro* the amount of APOA2-AT-specific protein sequences OR the amount of APOA2-ATQ-specific protein sequences allow to discriminate the pancreatic cystic tumor to be benign OR to be malignant on the basis of the respective amount, and thereby providing a significant contribution to the prior art for assisting in the determination of malignant pancreatic cystic tumor in the test subject.

As noted, D1 to D4 at best allow to determine whether a malignant pancreatic cystic tumor is present in a test subject. But these documents do not show beyond, and in particular they do not allow to discriminate between a benign or a malignant type of pancreatic cystic tumor. That is, the surprising finding of the present invention cannot be derived in an obvious manner by the skilled person from any one of D1 to D4.

5.4 It is therefore submitted that the subject-matter of claim 1 does involve an inventive step within the meaning of Art. 56 EPC.

5.5 The same observations likewise apply to the subject-matter of claim 9 as amended.

5.6 With respect to claim 13 (original claim 14), it is noted that this subject-matter provides a specific anti-APOA2-AT monoclonal antibody that specifically binds to the carboxyl-terminal region of the APOA2-AT protein (with the specified heavy chain CDRs and the light chain CDRs), and thus are highly useful tools for carrying out the method according to claim 1. Therefore, the subject-matter of amended claim 13 also does have an inventive merit in providing a contribution over the art, and thus is also inventive within the meaning of Art. 56 EPC.

6. It is thus assumed that claims are on file which meet the formal and substantive requirements of the EPC. The issuance of the Rule 71(3) Communication is thus respectfully requested. Merely for the sake of precaution, oral proceedings are requested as an auxiliary measure.

Respectfully submitted



Professional Representative

Enclosures
new claims 1 to 13 (clean and marked-up versions)